

Day-11

DATE:20-08-25

NAME: NANCY M

Today's task

1. To the existing books table, add additional column called price and add at least ten entries with some price values.

SQLQuery5.sql - P...ODA\nancy.m (107) X SQLQuery4.sql - P...YODA\nancy.m (56)

```
SELECT TOP (1000) [Id]
      ,[Title]
      ,[Description]
      ,[Rating]
      ,[Price]
FROM [demo_nancy].[dbo].[Books]
```

100 %

Results Messages

	Id	Title	Description	Rating	Price
1	5	Atomic Habits	Habits, Growth, Success	5	250
2	6	The Alchemist	Dreams, Journey, Destiny	4	280
3	7	To Kill a Mockingbird	Justice, Innocence, Racism	3	380
4	8	Sapiens	History, Evolution, Humanity	5	180
5	9	The Power of Now	Mindfulness, Presence, Awareness	2	780
6	10	The Great Gatsby	Love, Wealth, Illusion	4	680
7	11	Rich Dad Poor Dad	Money, Mindset, Freedom	3	280
8	12	Think and Grow Rich	Success, Desire, Belief	4	580
9	13	Pride and Prejudice	Love, Society, Class	5	80
10	14	The Silent Patient	Mystery, Trauma, Silence	2	10

1.1 Create a post request 'GetBooksOrderedByPrice()' with a boolean parameter

1.1.1 when parameter is true, then sort by price ascending order

1.1.2 when parameter is false, then sort by price descending order

```
[HttpPost("GetBooksOrderedByPrice")]
```

```
public IActionResult GetBooksOrderedByPrice([FromQuery] bool ascending)
```

```
{
```

```
    var books = ascending
```

```
        ? appDbContext.Books.OrderBy(b => b.Price).ToList()
```

```
        : appDbContext.Books.OrderByDescending(b => b.Price).ToList();
```

```
    return Ok(books);
```

```
}
```

The screenshot shows a REST client interface with a 'Parameters' tab. It displays a POST request to 'http://localhost:5246/Book/GetBooksOrderedByPrice?ascending=true'. The parameter 'ascending' is a boolean query parameter set to 'true'. Below the parameters, there are 'Execute' and 'Clear' buttons. The 'Responses' tab is also visible, showing the curl command and the request URL.

Name	Description
ascending	boolean (query)

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:5246/Book/GetBooksOrderedByPrice?ascending=true' \
  -H 'accept: */*' \
  -d ''
```

Request URL

```
http://localhost:5246/Book/GetBooksOrderedByPrice?ascending=true
```

Server response

Code Details

```
[
```

```
{
```

```
  "id": 14,
```

```
  "title": "The Silent Patient",
```

```
  "description": "Mystery, Trauma, Silence",
```

```
  "rating": 2,
```

```
  "price": 10
```

},

{

"id": 13,

"title": "Pride and Prejudice",

"description": "Love, Society, Class",

"rating": 5,

"price": 80

},

{

"id": 8,

"title": "Sapiens",

"description": "History, Evolution, Humanity",

"rating": 5,

"price": 180

},

{

"id": 5,

"title": "Atomic Habits",

"description": "Habits, Growth, Success",

"rating": 5,

"price": 250

},

{

"id": 6,

"title": "The Alchemist",
"description": "Dreams, Journey, Destiny",
"rating": 4,
"price": 280

},

{

"id": 11,
"title": "Rich Dad Poor Dad",
"description": "Money, Mindset, Freedom",
"rating": 3,
"price": 280

},

{

"id": 7,
"title": "To Kill a Mockingbird",
"description": "Justice, Innocence, Racism",
"rating": 3,
"price": 380

},

{

"id": 12,
"title": "Think and Grow Rich",
"description": "Success, Desire, Belief",
"rating": 4,

```
"price": 580
},
{
  "id": 10,
  "title": "The Great Gatsby",
  "description": "Love, Wealth, Illusion",
  "rating": 4,
  "price": 680
},
{
  "id": 9,
  "title": "The Power of Now",
  "description": "Mindfulness, Presence, Awareness",
  "rating": 2,
  "price": 780
}
]
```

Parameters

Cancel

Name	Description
ascending	false
boolean	
(query)	

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:5246/Book/GetBooksOrderedByPrice?ascending=false' \
  -H 'accept: */*' \
  -d ''
```

Request URL

```
http://localhost:5246/Book/GetBooksOrderedByPrice?ascending=false
```

Server response

Code	Details
200	Response body

```
[
{
  "id": 9,
  "title": "The Power of Now",
  "description": "Mindfulness, Presence, Awareness",
  "rating": 2,
  "price": 780
},
{
  "id": 10,
  "title": "The Great Gatsby",
  "description": "Love, Wealth, Illusion",
  "rating": 4,
  "price": 680
}
```

```
},  
  
{  
  "id": 12,  
  "title": "Think and Grow Rich",  
  "description": "Success, Desire, Belief",  
  "rating": 4,  
  "price": 580  
},  
  
{  
  "id": 7,  
  "title": "To Kill a Mockingbird",  
  "description": "Justice, Innocence, Racism",  
  "rating": 3,  
  "price": 380  
},  
  
{  
  "id": 6,  
  "title": "The Alchemist",  
  "description": "Dreams, Journey, Destiny",  
  "rating": 4,  
  "price": 280  
},  
  
{  
  "id": 11,
```

```
"title": "Rich Dad Poor Dad",  
"description": "Money, Mindset, Freedom",  
"rating": 3,  
"price": 280
```

```
},
```

```
{
```

```
"id": 5,  
"title": "Atomic Habits",  
"description": "Habits, Growth, Success",  
"rating": 5,  
"price": 250
```

```
},
```

```
{
```

```
"id": 8,  
"title": "Sapiens",  
"description": "History, Evolution, Humanity",  
"rating": 5,  
"price": 180
```

```
},
```

```
{
```

```
"id": 13,  
"title": "Pride and Prejudice",  
"description": "Love, Society, Class",  
"rating": 5,
```



```
"price": 80
},
{
  "id": 14,
  "title": "The Silent Patient",
  "description": "Mystery, Trauma, Silence",
  "rating": 2,
  "price": 10
}
]
```

1.2 Create a get request 'GetTop5Books()' which will return top 5 costliest books

```
[HttpGet("GetTop5Books")]
public IActionResult GetTop5Books()
{
    var topBooks = appDbContext.Books
        .OrderByDescending(b => b.Price)
        .Take(5)
        .ToList();
    return Ok(topBooks);
}
```

GET

/Book/GetTop5Books

⌵

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  https://localhost:5246/Book/GetTop5Books' \
  -H 'accept: */*'

```

Request URL

```
http://localhost:5246/Book/GetTop5Books

```

Server response

Code	Details
200	Response body <pre>[{ "id": 9, "title": "The Power of Now", "description": "Mindfulness, Presence, Awareness", "rating": 2, "price": 780 }, { "id": 10, "title": "The Great Gatsby", "description": "Love, Wealth, Illusion", "rating": 4, "price": 680 }, { "id": 12, "title": "Think and Grow Rich", "description": "Success, Desire, Belief", "rating": 4, "price": 580 }, { "id": 7, "title": "To Kill a Mockingbird", "description": "Justice, Innocence, Racism", "rating": 3, "price": 380 }] </pre>

```
[
  {
    "id": 9,
    "title": "The Power of Now",
    "description": "Mindfulness, Presence, Awareness",
    "rating": 2,
    "price": 780
  },
  {
    "id": 10,
    "title": "The Great Gatsby",
    "description": "Love, Wealth, Illusion",
    "rating": 4,
    "price": 680
  },
  {
    "id": 12,
    "title": "Think and Grow Rich",
    "description": "Success, Desire, Belief",
    "rating": 4,
    "price": 580
  },
  {
    "id": 7,
    "title": "To Kill a Mockingbird",
    "description": "Justice, Innocence, Racism",
    "rating": 3,
    "price": 380
  }
]
```

```

    },
    {
      "id": 6,
      "title": "The Alchemist",
      "description": "Dreams, Journey, Destiny",
      "rating": 4,
      "price": 280
    }
  ]

```

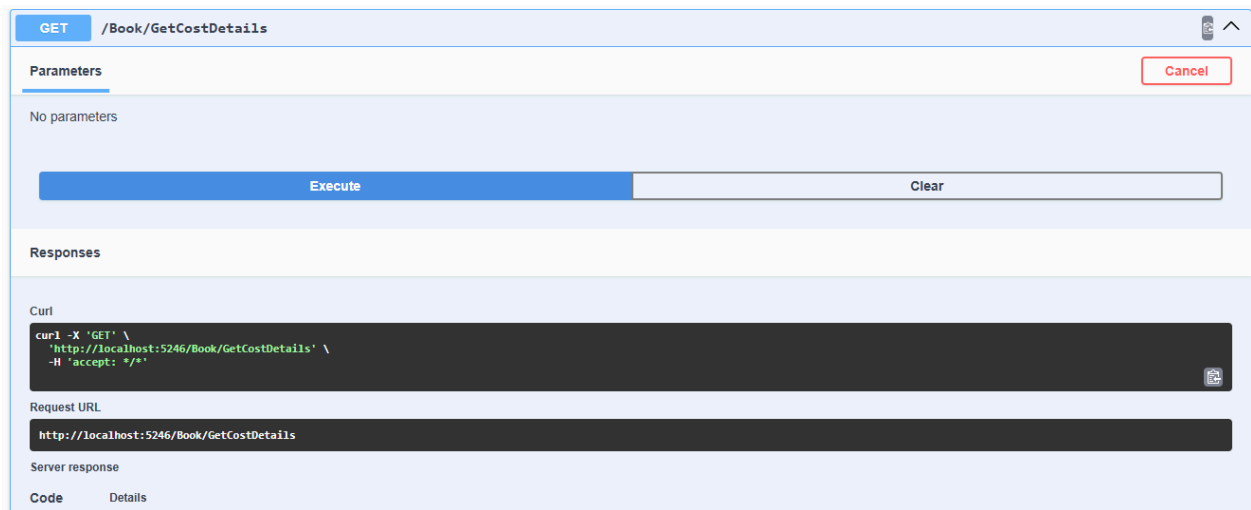
1.3 Create a get request 'GetCostDetails()' which will return the total price and average price of all books.

```

[HttpGet("GetCostDetails")]
public IActionResult GetCostDetails()
{
    var total = appDbContext.Books.Sum(b => b.Price);
    var average = appDbContext.Books.Average(b => b.Price);

    return Ok(new { TotalPrice = total, AveragePrice = average });
}

```



```

{
  "totalPrice": 3500,
  "averagePrice": 350
}

```

